B. Tech. Computer Engineering IV Sem. 1st Assignment – 2024

Paper Code: ***CEN-404*** **OPERATING SYSTEM**

S. No.	Questions
Q1.	What are the distinctions among logical, relative, and physical addresses?
Q2.	Explain the difference between internal and external fragmentation.
Q3.	Under what circumstances do page faults occur? Describe the actions taken by the operating system when a page fault occurs.
Q4.	A paging scheme uses a Translation Look-aside Buffer (TLB). A TLB-access takes 20 ns and the main memory access takes 200 ns. What is the effective access time (in ns) if the TLB hit ratio is 90% and there is no page-fault?
Q5.	What is compaction? What are its advantages and disadvantages of using compaction?
Q6.	Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB. These partitions need to be allocated to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order.
Q7.	Consider the following page reference string: 1, 2, 3, 4, 5, 2,3, 4, 1, 6, 7, 8, 7, 8, 9, 7, 8, 9, 5, 4, 5, 4, 2 How many page faults would occur for the following replacement algorithms, assuming four frames? Remember that all frames are initially empty, so your first unique pages will all cost one fault each. I. FIFO II. Optimal Page replacement III. LRU IV. Least Frequently Used
Q8.	Write Short notes on: I. Buddy System II. Hashed Page Table III. Inverted Page Table IV. Second-Chance Algorithm V. Belady's Anomaly
Q9.	What is Demand Paging? Discuss the hardware support required to support demand paging.
Q10.	What is the cause of thrashing? How does the system detect thrashing? Once it detects thrashing, what can the system do to eliminate this problem?